	GLOBAL ECONOMICS **Grade: 10th** **Learning Evidence 1** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2025–2026**

Learning objective: Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.	**Assessment criteria:** C1: Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision. C2: Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison. C3: Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison. C4: Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

Student's name: _____ **Group:** _____ **Date:** _____

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Question	C1	C2	C3	C4
1	1	2	–	–
2	1	3	–	–
3	2	5	–	–
4	1	–	2	–
5	1	–	3	–
6	2	–	5	–
7	–	–	–	2
8	–	–	–	2
9	–	–	–	2
Total	**8**	**10**	**10**	**6**

1. Meal Delivery Service [C1-1 C2-2]

A meal delivery company is planning its service offering for an upcoming operating period. A standard delivery service is expected to generate profits of twenty thousand US dollars under high demand and twelve thousand US dollars under low demand, while a premium delivery service is expected to generate thirty-four thousand US dollars and seven thousand US dollars, respectively. The company must select its delivery service before the actual level of customer demand is known. Characterize the key elements of the decision-making situation and construct the payoff table.

2. Small Coffee Roastery Production [C1-1 C2-3]

A small coffee roastery is planning its production capacity for an upcoming sales period. A standard roasting machine can generate 4 thousand BRL per production batch, while an industrial roasting machine can generate 6 thousand BRL per batch. Based on expected demand, the standard roasting machine would complete 18 batches under high demand or 12 under low demand, whereas the industrial roasting machine would complete 15 or 10 batches, respectively. In addition, the roastery would receive a fixed sales bonus of 7 thousand BRL if demand is high and incur a fixed loss of 4 thousand BRL if demand is low. Characterize the key elements of the decision-making situation and construct the payoff table.

3. Online Fitness Membership [C1-2 C2-5]

An online fitness company must choose between a standard membership and a flexible membership before customer demand is known. Demand may be high or low. The standard membership generates 5 thousand BRL per active member and also provides a fixed operating contribution of 5 thousand BRL. The flexible membership generates 7 thousand BRL per active member and provides a fixed operating contribution of 8 thousand BRL. High demand adds another 3 thousand BRL per active member to either membership, while low demand adds 1 thousand BRL per active member. The standard membership is expected to have 15 active members under high demand and 9 under low demand; the flexible membership is expected to have 12 active members under high demand and 7 under low demand. Characterize the key elements of the decision-making situation and construct the payoff table.

4. Neighborhood Bakery Expansion [C1-1 C3-2]

A neighborhood bakery must decide whether to introduce a breakfast menu or maintain its current product selection before customer traffic is known. Traffic may be strong or weak. All values are in million BRL. Introducing the breakfast menu produces revenue of 68 under strong traffic and 38 under weak traffic, while maintaining the current selection produces revenue of 59 under strong traffic and 49 under weak traffic. The corresponding costs are 23 and 21 for the breakfast menu under strong and weak traffic, respectively, and 24 and 22 for the current selection under strong and weak traffic, respectively. Characterize the key elements of the decision-making situation and construct the payoff table.

5. Bakery Product Production [C1-1 C3-3]

A bakery is planning its production for an upcoming holiday season. Revenue per production batch is 20 thousand EUR for cakes and 16 thousand EUR for pastries. Under high customer demand, the bakery expects to produce 9 batches of cakes and 6 batches of pastries, while under low customer demand it expects to produce 5 batches of cakes and 4 batches of pastries. The cost per production batch is 8 thousand EUR under high customer demand and 11 thousand EUR under low customer demand. Characterize the key elements of the decision-making situation and construct the payoff table.

6. Meal Delivery Service Packages [C1-2 C3-5]

A meal delivery service must choose between a standard meal package and a premium meal package before weekly demand is known. Demand may be high or low. Quantities, in hundreds of meal packages, are 10 and 7 for the standard package and 8 and 5 for the premium package under high and low demand, respectively. The package-based revenue rates are 7 and 12 thousand USD per hundred meal packages for the standard and premium packages. High demand adds a state-based revenue rate of 3 thousand USD per hundred meal packages; low demand adds 1. The package-based cost rates are 3 and 5 thousand USD per hundred meal packages, respectively. The fixed operating costs are 14 thousand USD under high demand and 18 thousand USD under low demand. Characterize the key elements of the decision-making situation and construct the payoff table.

7. Solar Energy Investment Strategy **[C4-2]**

A renewable energy company must choose a strategy for investing in a new market. Energy demand could be high, medium, or low. All payoffs are in millions of dollars.

	Energy Demand		
Alternative	High	Medium	Low
Utility-scale solar farm	82	38	6
Rooftop solar program	76	51	34
Community solar project	97	44	26
Postpone investment	28	28	28

What would the optimistic decision be? What would the conservative decision be? Justify your answer.

8. Restaurant Expansion Strategy **[C4-2]**

A restaurant company must choose a strategy for expanding a new location. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Customer demand		
Alternative	High	Medium	Low
Large restaurant	78	36	12
Medium restaurant	68	49	24
Small restaurant	91	47	29
No expansion	61	42	37

What would the optimistic decision be? What would the conservative decision be? Justify your answer.

9. Renewable Energy Investment Strategy **[C4-2]**

A renewable energy company must choose a strategy for introducing a new solar power system. Energy market conditions could be highly favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

	Energy market conditions			
Alternative	Highly favourable	Favourable	Stable	Unfavourable
Full-scale solar	132	61	10	-38
Expanded solar	91	67	32	6
Basic solar	65	54	40	22
Solar leasing	51	46	37	30

What would the optimistic decision be? What would the conservative decision be? Justify your answer.